Object Oriented Programming

LAB-03 Solutions

Task 01 - Part 1

```
#include <iostream>
#include <string>
#include <iomanip>
#include <cstdlib>
#include <ctime>
using namespace std;

struct InventoryItem
{
    int code;
    string name;
    double cost;
    double sellingPrice;
};

// takes name and cost from user, generates code and selling price (40% profit)
InventoryItem insertItem()
{
    InventoryItem item;

    item.code = rand() % 9000 + 1000;   // random number between 1000 and 9999

    cin.ignore();
    cout << "Enter item name: ";
    getline(cin, item.name);

    cout << "Enter item cost: ";
    cin >> item.cost;

    item.sellingPrice = item.cost + item.cost * 0.40;

    return item;
}

// displays one item in a row
void displayStock(const InventoryItem* item)
{
    cout << left << setw(10) << item->code
         << setw(20) << item->name
         << setw(12) << item->cost
         << setw(12) << item->sellingPrice << endl;
}

int main()
{
    srand(time(0));

    const int SIZE = 10;
    InventoryItem stock[SIZE];

    for (int i = 0; i < SIZE; i++)
    {
        cout << "\nItem " << i + 1 << endl;
        stock[i] = insertItem();
    }

    cout << "\n---------------- STOCK ----------------" << endl;
    cout << left << setw(10) << "Code"
         << setw(20) << "Name"
         << setw(12) << "Cost"
         << setw(12) << "Price" << endl;
    cout << "-------------------------------------------------" << endl;

    for (int i = 0; i < SIZE; i++)
    {
        displayStock(&stock[i]);
    }

    return 0;
}
```

Task 01 - Part 2

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;

struct InventoryItem
{
    string code;
    string name;
    double cost;
    double sellingPrice;
};

int counter = 31203;   // running number used in the item code

// takes name and cost from user, generates code and selling price (40% profit)
InventoryItem insertItem()
{
    InventoryItem item;

    cin.ignore();
    cout << "Enter item name: ";
    getline(cin, item.name);

    cout << "Enter item cost: ";
    cin >> item.cost;

    // first 3 letters of the name in upper case
    string prefix = item.name.substr(0, 3);
    for (int i = 0; i < prefix.length(); i++)
    {
        prefix[i] = toupper(prefix[i]);
    }

    // code = PREFIX_number  (example: COM_31203)
    item.code = prefix + "_" + to_string(counter);
    counter++;

    item.sellingPrice = item.cost + item.cost * 0.40;

    return item;
}

// displays one item in a row
void displayStock(const InventoryItem* item)
{
    cout << left << setw(12) << item->code
         << setw(20) << item->name
         << setw(12) << item->cost
         << setw(12) << item->sellingPrice << endl;
}

int main()
{
    const int SIZE = 10;
    InventoryItem stock[SIZE];

    for (int i = 0; i < SIZE; i++)
    {
        cout << "\nItem " << i + 1 << endl;
        stock[i] = insertItem();
    }

    cout << "\n---------------- STOCK ----------------" << endl;
    cout << left << setw(12) << "Code"
         << setw(20) << "Name"
         << setw(12) << "Cost"
         << setw(12) << "Price" << endl;
    cout << "-------------------------------------------------" << endl;

    for (int i = 0; i < SIZE; i++)
    {
        displayStock(&stock[i]);
    }

    return 0;
}
```

Task 02

```
#include <iostream>
using namespace std;

struct FeetInches
{
    int feet;
    int inches;
};

struct RoomDimension
{
    FeetInches length;
    FeetInches Width;
};

const int PRICE_PER_SQ_FOOT = 18;   // Rs. per square foot

// calculates the carpeting cost of one room
int computeTheCarpetingCost(const RoomDimension& room)
{
    // convert feet + inches into feet
    double length = room.length.feet + room.length.inches / 12.0;
    double width = room.Width.feet + room.Width.inches / 12.0;

    double area = length * width;

    return (int)(area * PRICE_PER_SQ_FOOT);
}

int main()
{
    int numRooms;
    cout << "Enter the number of rooms in the house: ";
    cin >> numRooms;

    // dynamic array of rooms
    RoomDimension* rooms = new RoomDimension[numRooms];

    for (int i = 0; i < numRooms; i++)
    {
        cout << "\nRoom " << i + 1 << endl;
        cout << "Enter length (feet inches): ";
        cin >> rooms[i].length.feet >> rooms[i].length.inches;
        cout << "Enter width (feet inches): ";
        cin >> rooms[i].Width.feet >> rooms[i].Width.inches;
    }

    int choice;
    do
    {
        cout << "\n----------- MENU -----------" << endl;
        cout << "1. Total cost of carpeting the house" << endl;
        cout << "2. Cost of carpeting a specific room" << endl;
        cout << "3. Exit" << endl;
        cout << "Enter your choice: ";
        cin >> choice;

        if (choice == 1)
        {
            int total = 0;
            for (int i = 0; i < numRooms; i++)
            {
                total += computeTheCarpetingCost(rooms[i]);
            }
            cout << "Total cost = Rs. " << total << endl;
        }
        else if (choice == 2)
        {
            int roomNo;
            cout << "Enter room number (1 to " << numRooms << "): ";
            cin >> roomNo;

            if (roomNo >= 1 && roomNo <= numRooms)
            {
                cout << "Cost of room " << roomNo << " = Rs. "
                     << computeTheCarpetingCost(rooms[roomNo - 1]) << endl;
            }
            else
```

```
            {
                cout << "Invalid room number!" << endl;
            }
        }
        else if (choice != 3)
        {
            cout << "Invalid choice, try again." << endl;
        }

    } while (choice != 3);

    // free memory
    delete[] rooms;

    return 0;
}
```